

ORIGINAL ARTICLE

Engaging Customers, Embracing AI: How Suppliers Respond to Trade Tensions and Improve Performance

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ABSTRACT

Academic Summary: Firms face intensifying pressure to innovate amid escalating US–China trade tensions. This study investigates how they integrate artificial intelligence (AI) to respond to such disruptions. We reconceptualize induced innovation theory as adversity-induced innovation theory and test hypotheses using data from listed Chinese manufacturing firms, WTO tariff records, and customs data (2015–2022). Results indicate that higher tariff exposure increases AI integration, particularly among firms with stronger customer engagement needs. AI integration improves supply chain efficiency, sales performance, and market value. However, performance benefits follow an inverted U-shape: extreme tariff exposure diminishes AI's compensatory effects. The study extends induced innovation theory to geopolitical adversities and advances stakeholder engagement theory by identifying customer engagement as a boundary condition for technology adoption. We establish causal relationships between tariff-induced AI integration and firm performance through instrumental variable analysis and robustness checks. Findings inform how firms can strategically use technological innovation and customer collaboration to strengthen resilience under external shocks. Managers should view trade tensions as catalysts for AI integration rather than merely as obstacles.

Managerial Summary: Firms facing trade tensions should view them as catalysts for strategic AI integration rather than merely as obstacles. Our research shows that firms experiencing higher tariff exposure that invest in AI achieve measurable improvements in supply chain efficiency, sales performance, and market valuations. However, AI's compensatory benefits diminish at extreme tariff levels, revealing an optimal investment threshold that managers should consider. The benefits of AI integration are particularly strong for firms with high customer engagement needs, such as those with concentrated customer bases, substantial international sales, or significant customer-specific investments. To maximize returns, managers should prioritize AI technologies that strengthen customer relationships by enhancing communication, responsiveness, and service personalization during trade disruptions. Successful implementation requires navigating technical and organizational challenges, including talent acquisition, data quality concerns, and integration complexities. Despite these hurdles, our findings confirm AI's strategic value as a compensatory innovation tool that not only mitigates tariff impacts but also transforms supplier-customer relationships. By aligning AI initiatives with customer engagement objectives, firms can turn geopolitical adversity into a competitive advantage. Companies that integrate AI strategically can maintain operational continuity, preserve market position, and emerge stronger from trade disruptions.

[Correction added on 11 February 2026 after first online publication: The copyright line was changed.]

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1 | Introduction

The resurgence of nationalism and protectionism has become a defining feature of today's global economy, fueling trade tensions and global supply chain disruptions (Chor and Li 2024; Grossman et al. 2024). Geopolitical flashpoints, including the US–China trade dispute and conflicts in Ukraine and Gaza, have forced firms to overhaul supply networks, disrupting the movement of goods, capital, and information on a global scale (Moradlou et al. 2021). Tariffs, as a central instrument of these changes, have undermined the stability and resilience of supply chains (Benguria et al. 2022), contributing to sharp declines in investment, production, R&D, and profitability (Benguria et al. 2022; Chor and Li 2024). In response, firms have begun to diversify suppliers and adopt new technologies to mitigate risk and restore operational continuity (Dong and Kouvelis 2020).

The integration of Artificial Intelligence (AI) into global supply chain management has become a key strategic approach for addressing geopolitical disruptions and generating value for diverse stakeholder groups (Haenlein and Kaplan 2019). AI technologies, including machine learning, natural language processing, computer vision, and predictive analytics, enable firms to analyze vast amounts of data from various sources in real time, providing unprecedented insights and agility (Davenport and Ronanki 2018). AI-powered innovations could be utilized to enhance risk management, support decision-making, increase efficiency, improve customer engagement, and foster innovation (Baryannis et al. 2019; Haenlein and Kaplan 2019; Toorajipour et al. 2021). However, the integration of AI is not without its challenges, which include addressing talent shortages, managing ethical concerns, ensuring data quality, and overcoming integration difficulties (Wamba et al. 2020).

AI integration becomes increasingly important as firms respond to shifts in trade policy and adjust their supply chain structures. However, in light of the implementation challenges, successful adoption often depends not only on internal technological capacity but also on the degree to which firms align these technologies with the expectations of external stakeholders, particularly primary customers. This alignment matters because suppliers' ability to maintain effective engagement with key business customers can significantly influence both the implementation of AI and resulting performance outcomes (Brodie et al. 2011). Customer engagement involves several interrelated elements: the strength of collaboration, mutual understanding, and trust between suppliers and customers (Brodie et al. 2011); the extent to which AI-enabled systems and offerings are adapted to meet specific customer needs (Tuli et al. 2007); and the integration of customer priorities into suppliers' operational decisions, strategic initiatives, and value delivery processes (Vivek et al. 2012).

According to the induced innovation theory (Acemoglu 2010; Hicks 1932), changes in relative factor prices steer the trajectory of technological change. This theory has been empirically supported across various sectors, including energy (Newell et al. 2010; Popp 2002), agriculture (Ruttan 1977; Olmstead and Rhode 1993), and environmental technologies (Ren et al. 2022). Recent studies have extended its application to explain how economic challenges drive innovation, such as emissions trading systems stimulating green innovation (Ren et al. 2022) and

rising labor costs driving technological advancements in China's secondary industry (Fleisher et al. 2021). Despite this rich body of literature, the existing literature on induced innovation theory fails to adequately address its application to suppliers' responses to trade disruptions, particularly in the context of AI integration.

Current research has not fully explored how sudden changes in trade policy, particularly the introduction of higher tariffs, can push suppliers to quickly adopt new technologies. Moreover, there is a lack of empirical evidence linking AI integration to firm performance in times of trade disruptions. Specifically, the mechanisms by which tariff hikes might induce AI integration in supply chain management are not well understood (Acemoglu 2015). This gap is particularly problematic given the increasing frequency and impact of trade disruptions in today's global economy (Benguria et al. 2022).

To address these gaps in the innovation literature, this study aims to examine whether firms espouse AI integration as a strategic response to geopolitical disruptions, specifically, the tariff shock, and how the firm's need for engagement with its primary customers plays a role in this response. Extending the induced innovation theory to the broader area of adversity, we argue that in response to U.S. tariff hikes, Chinese suppliers are likely to turn to AI technological integration to offset the increased costs. We hypothesize that firms with higher exposure to tariff increases are more likely to embrace AI integration in an effort to mitigate the impact of trade policy uncertainty on their operations and performance. Moreover, we posit that this relationship is stronger for firms with higher needs for engagement with their primary customers. Finally, we propose that for firms with higher tariff exposure, a higher level of AI integration leads to improved firm performance.

Our study makes several important contributions to the growing literature on AI, stakeholder engagement, and innovation management. First, we extend traditional induced innovation theory to broader adversity-induced innovation and provide empirical evidence of the strategic value of AI integration in mitigating the negative effects of tariff exposure. Second, we introduce primary customer engagement as a key moderator and explore its role in influencing firms' AI integration responses to such adversity. Finally, the study enriches the literature on AI's strategic value by providing empirical evidence linking AI integration to firm performance during trade disruptions.

2 | Theory and Hypotheses

2.1 | Induced Innovation Theory

Induced innovation theory, first proposed by Hicks (1932) and further developed by Acemoglu (2010), posits that shifts in relative factor prices steer the trajectory of innovation, as firms are incentivized to develop and adopt technologies that reduce their reliance on more expensive inputs (Acemoglu 2015). The theory finds support from various studies. For example, higher energy prices have led to increased innovation in energy-efficient technologies (Popp 2002), and energy efficiency regulations have induced innovation in air conditioners and water

heaters (Newell et al. 2010). The theory has also explained technological and institutional changes in agriculture (Asche and Smith 2018; Olmstead and Rhode 1993; Ruttan 1977) and energy sectors (Grubb et al. 2020). Recent applications demonstrate how economic challenges drive technological change: emissions trading systems stimulate green innovation (Ren et al. 2022), financial constraints enhance R&D efficiency in green technologies (Zhang and Jin 2021), and rising labor costs drive innovation in China's secondary industry (Fleisher et al. 2021).

Building on these findings, we argue that induced innovation theory can be extended to account for a wider range of adversities beyond purely economic conditions. Adversity refers to circumstances that hinder individuals, firms, or governments in pursuing their objectives (Asay and Bair 2021; Rosenzweig and Grinstein 2016). The COVID-19 pandemic illustrates this point: severe global disruption stimulated significant innovation across healthcare, education, and food services, leading to the development of new solutions to address pandemic-related challenges (Asay and Bair 2021). Both the Western proverb "Necessity is the mother of invention" and the Chinese proverb "In hardship, consider change" reflect this principle, suggesting a cross-cultural recognition that difficult circumstances often encourage creative problem-solving. However, adversity has boundaries. Extreme or prolonged hardship, such as persistent poverty, may suppress creativity and innovation (Bair 2020), indicating a threshold beyond which adversity becomes detrimental. Access to resources, supportive institutional arrangements, and organizational capabilities strongly influence whether adversity fosters innovation. This broader view of adversity-induced innovation provides opportunities for examining how different forms of challenge influence technological development across sectors and societies.

2.2 | The Rise of AI and Firms' AI Integration

AI has become a key driver of business innovation, transforming how companies create value across industries (Haenlein and Kaplan 2019; Hollebeek et al. 2021; Wetzels 2021). We define AI integration as the adoption, investment in, and strategic implementation of AI and related technologies to address external challenges and influence performance. This includes cognitive AI, which performs human-like perception, reasoning, and interaction through natural language processing, speech recognition, and computer vision, and analytical AI, which examines data for insights, predictions, and decision support using machine learning, deep learning, and predictive analytics. Companies may employ various AI technologies to address trade disruptions and create stakeholder value.

AI offers substantial economic benefits. It analyzes extensive, real-time data on business processes and product usage (Rammer et al. 2022). Firms can automate tasks, improve operations, and enhance product performance. AI generates insights, predictions, and automated decision-making capabilities (Baryannis et al. 2019; Toorajipour et al. 2021). It supports data-driven business models and helps develop new products and services (Lee et al. 2019). AI adoption is linked to revenue growth, particularly when combined with complementary technologies

(Ardito et al. 2024). It also transforms customer interactions through behavioral insights and personalized marketing (Filieri et al. 2021; Kopalle et al. 2022). Beyond economic gains, AI provides social and sustainability benefits. It helps firms address sustainability challenges, manage waste, and facilitate sustainable transformations in production and innovation (Mariani et al. 2023). AI supports cost reduction, enhances productivity, shortens product development cycles, and improves decision-making (Verganti et al. 2020).

However, AI integration presents significant challenges. Concerns about job loss from automation (Frey and Osborne 2017), the time required for process adaptation and worker reskilling (Tambe et al. 2019), and issues related to data quality, skill gaps, technological uncertainty, and regulations all hinder adoption (Haefner et al. 2021). To realize AI's benefits fully, firms must move beyond mere adoption. They should integrate AI into strategy and operations (Raisch and Krakowski 2021). This requires strategic alignment, resource allocation, and organizational integration (Haefner et al. 2021), employee capabilities (Garbuio and Lin 2021), and external innovation partnerships (Sjödin et al. 2021). Our conceptualization of AI integration allows for assessment of how organizations leverage AI to address external challenges and how such integration influences performance.

2.3 | Customer Engagement

Supply chain disruptions from trade tensions create unique challenges for firms that depend heavily on their primary customers. How these firms respond to adversity may depend on the strength and nature of their customer relationships. Understanding this connection requires examining customer engagement as a strategic construct. Customer engagement involves resource investments in interactions, activities, and relationships that extend beyond simple exchanges (Vivek et al. 2012). Drawing on stakeholder theory and service-dominant logic, engagement research recognizes that firms must consider diverse stakeholder interests to create value (Freeman et al. 2010; Hollebeek et al. 2022). Customer engagement specifically includes knowledge sharing (Filieri 2013), cooperation, and collaboration toward mutual objectives (Viglia et al. 2018, 2023). These collaborative processes require substantive resource commitment from both parties (Hollebeek et al. 2023).

In this study, we focus on the need for primary customer engagement, defined as the extent to which a supplier's operations, strategies, and value creation processes depend on close collaboration, mutual understanding, and trust with its primary customers (Brodie et al. 2011; Vivek et al. 2012). This need intensifies when suppliers make significant relationship-specific investments and customize their offerings to meet the distinct requirements of their major customers (Tuli et al. 2007). Suppliers with high dependence on primary customer engagement face different pressures and opportunities when responding to trade disruptions compared with suppliers with more transactional customer relationships.

The need for primary customer engagement becomes particularly relevant in the context of AI integration. Suppliers with

strong customer engagement relationships may face pressure to adopt AI technologies that align with customer expectations or requirements. These suppliers may also have access to richer customer data and collaborative opportunities that enable more effective AI implementation. Conversely, the costs and risks of AI adoption may be higher for suppliers whose primary customers demand stability and proven solutions rather than technological experimentation. Understanding how the need for primary customer engagement influences AI integration decisions under trade adversity is therefore critical.

Table 1 summarizes research from trade economics, supply chain management, and customer engagement fields, showing how our study integrates these perspectives to examine AI integration under trade tensions. Building on this foundation, we develop adversity-induced innovation theory to explain how the interaction between trade disruptions and customer engagement needs affects firm AI integration.

2.4 | Hypothesis Development

Building on our extended adversity-induced innovation theory, we propose that Chinese firms with high exposure to tariffs are likely to seek technological solutions to mitigate the impact of increased costs. AI represents a particularly promising avenue for such innovation. These technologies can automate tasks, replace labor, and enhance productivity (Acemoglu and Restrepo 2018), enabling firms to offset tariff burdens and maintain their competitive position. AI serves as a key tool for achieving sustainable supply chain objectives, enhancing environmentally conscious practices and optimizing supplier selection, forecasting, and logistics (Hao and Demir 2025). It supports supplier selection through data-driven decision-making, transforms procurement via automation (Nissen and Sengupta 2006), improves warehousing and production processes (Yang et al. 2022), and strengthens demand forecasting (Simchi-Levi and Wu 2018). Thus, we hypothesize:

H1. *Suppliers with higher exposure to tariffs will exhibit a greater degree of AI integration.*

Firms with strong primary customer engagement possess both the motivation and the capability to respond strategically to tariff pressures through AI integration. As established in Section 2.3, deep collaboration with key customers maintains communication channels and provides insights into their requirements (Ganesan et al. 2005; Rindfleisch and Moorman 2001). This engagement becomes particularly valuable under tariff pressure, enabling firms to interpret market signals and adjust to tariff-induced demand shifts (Tuli et al. 2007). Trust developed through these relationships fosters an environment conducive to focused innovation efforts (Rindfleisch and Moorman 2003).

The exchange of complex information through customer engagement (Ganesan et al. 2005) helps firms understand how tariffs affect customer needs, aligning with induced innovation theory's principle that price signals guide technological change. This reduced uncertainty allows firms to concentrate on value creation through AI rather than on negotiation. Deep engagement establishes a foundation for joint problem-solving, and

firms leverage their understanding of customer needs (Vivek et al. 2012) to develop AI solutions that address tariff challenges. The combination of tariff-induced price pressures and the need for superior market understanding through primary customer engagement creates a strong impetus for AI integration. Based on these considerations, we hypothesize:

H2. *The positive relationship between tariff exposure and AI integration is stronger for supplier firms with a greater need for primary customer engagement.*

When facing tariff exposure, firms motivated to adopt AI technologies must address several implementation challenges, including data quality issues (Hazen et al. 2014), privacy concerns (Martin and Murphy 2017), difficulty acquiring AI expertise, organizational resistance, ethical considerations, and high implementation costs (Moktadir et al. 2019). However, induced innovation theory suggests that firms with higher tariff exposure have greater incentives to pursue AI integration, potentially improving performance despite these obstacles.

AI integration can generate measurable benefits across multiple business operations. In supply chains, AI optimizes costs by identifying inefficiencies (Gunasekaran et al. 2017) and enhances forecasting, inventory management, and production planning (Dubey et al. 2021). In sales, AI provides a comprehensive view of the supply chain and highlights potential risks (Dubey et al. 2021). For innovation and firm value, AI enables product, service, and process improvements (Zhan and Tan 2020), which can offset the impact of tariffs. Additionally, AI supports decision-making with real-time data (Gunasekaran et al. 2017), enhances operational efficiency for improved profitability (Hazen et al. 2016), and mitigates supply chain risks for more stable valuations. Consequently, tariff exposure can act as a catalyst for AI integration, which in turn strengthens firm performance. Thus, we hypothesize:

H3. *The higher the degree of exposure to tariffs, the greater the positive impact of AI technological integration on firm performance, as measured by (a) supply chain performance, (b) sales performance, and (c) firm value.*

Figure 1 illustrates our proposed research model.

3 | Methodology

3.1 | Sample Firms and U.S. Tariff Exposure

This study uses a large sample of manufacturing firms listed on China's Shanghai and Shenzhen stock exchanges from 2015 to 2022. Financial data are obtained from the Chinese Stock Market and Accounting Research Database (CSMAR). Our analysis period begins March 22, 2018, when the United States announced tariff increases on \$50 billion of Chinese imports (Benguria et al. 2022; Chor and Li 2024; Chen et al. 2023; Huang et al. 2023). Tariff data, including U.S. import tariffs on China and Chinese retaliatory tariffs, are collected from Li (2018) and Most Favored Nation (MFN) tariffs from the WTO Tariff Download Facility. Firm-level trade data come from the China Customs Dataset (2013–2016).

TABLE 1 | Review of relevant studies.

Authors (year)	Research focus	Key findings
Induced innovation theory		
Hicks (1932)	Factor prices and technological change	Changes in relative factor prices direct innovations toward saving the more expensive factor; firms develop technologies that reduce reliance on costlier inputs.
Acemoglu (2010)	Labor scarcity and innovation	Labor scarcity encourages labor-saving technological development under specific economic conditions, identified when labor scarcity promotes rather than hinders innovation.
Popp (2002)	Energy prices driving innovation	Higher energy prices increased energy-efficient technology innovation; demonstrated market signals effectively redirected R&D efforts.
Newell et al. (2010)	Regulatory effects on technology	Energy efficiency regulations induced specific innovations in air conditioners and water heaters, showing the combined influence of market forces and government policies.
Ruttan (1977)	Agricultural innovation patterns	Resource constraints and factor prices prompted agricultural technology development, linking technical innovation with institutional changes in agricultural systems.
Fleisher et al. (2021)	Innovation in China's industry	Rising labor costs in the secondary industry drove specific technological adaptations; firms responded to changing factor prices with labor-saving innovations.
Ren et al. (2022)	Environmental policy and innovation	Emissions trading systems stimulated green innovation; policy-induced price changes effectively redirected innovation toward environmental solutions.
Tariffs and firm performance		
Benguria et al. (2022)	Tariffs' impact on Chinese firms	Firms dealing with greater trade policy uncertainty cut back on investment, research and development, and saw their profits decline.
Chor and Li (2024)	US-China tariff war on China's economy	Increases in US tariff exposure reduced economic activity; the most highly exposed portions of the population experienced decreased income per capita.
Fan et al. (2022)	Trade war impact on US firms with Chinese suppliers	US firms with direct Chinese suppliers showed worse inventory management and profitability; effects worsened with higher outsourcing and supply base complexity.
Amiti et al. (2019)	Tariffs' effects on prices and welfare	US tariffs passed through to domestic prices; the burden fell on US consumers, with no gains for domestic producers.
Flaen and Pierce (2019)	Tariffs and manufacturing	Tariffs reduced manufacturing employment and increased producer prices; rising input costs outweighed potential benefits from reduced competition.
Handley et al. (2020)	Firm-level trade policy uncertainty exposure	Trade policy uncertainty exposure had negative impacts on investment and market valuation for more exposed firms.
AI and supply chain management		
Toorajipour et al. (2021)	AI techniques in supply chain management	Mapped current and potential AI applications across logistics, marketing, supply chain, and production fields.
Baryannis et al. (2019)	AI for supply chain risk management	AI techniques can address risk identification, assessment, and response; they provide mapping of AI methodologies in supply chain risk management.
Hao and Demir (2025)	AI enablers and constraints in supply chains	Identified factors influencing AI integration across development stages; showed interactions between technological, organizational, and environmental factors.

(Continues)

TABLE 1 | (Continued)

Authors (year)	Research focus	Key findings
Babina et al. (2024)	AI impacts on firm growth	AI adoption increases firm growth and innovation capabilities; improves market insights and product development.
Belhadi et al. (2022)	AI for supply chain resilience	AI can structure operating rules for supply chain reengineering during disruptions through decision support systems.
Modgil et al. (2022)	AI and supply chain resilience during COVID-19	AI helped maintain resilience through scenario planning, automated risk monitoring, and visibility enhancement.
Customer engagement		
Hollebeek et al. (2022)	Stakeholder engagement theory	Framework based on stakeholder roles, mutual value creation process, and beyond-interaction outcomes.
Hollebeek et al. (2021)	Customer engagement with automated services	Positioned AI-customer interactions as engagement with active stakeholders rather than passive tools.
Viglia et al. (2023)	Multi-stakeholder engagement ecosystems	Explored value co-creation in multi-stakeholder ecosystems, applicable to supplier-customer relationships using AI.
Kumar et al. (2019)	Customer engagement across international markets	Customer engagement varies by cultural context; engagement strategies need adaptation for different markets.
Our study	Tariff-induced AI integration and firm performance	We examine how tariff exposure drives strategic AI integration, improving firm performance, with primary customer engagement moderating this relationship.

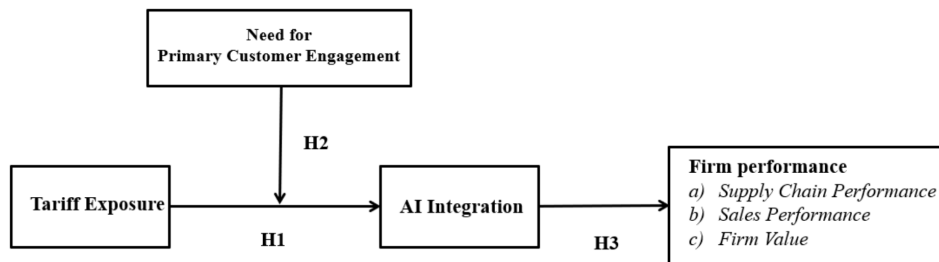


FIGURE 1 | Research model.

We combine tariff rates and firm-level customs data to construct the time-varying measures of firm-level tariff exposure of a Chinese firm i at time t as follows:

$$\text{Tariff}_{it}^{\text{U.S.}} = \sum_{j \in j_i^e} \left[\frac{X_{ij0}^{\text{U.S.}}}{\sum_{s \in j_i^e} X_{is0}^{\text{U.S.}}} \tau_{jt}^{\text{U.S.}} \right]$$

where $\tau_{jt}^{\text{U.S.}}$ represents good j ad valorem tariff at time t , $X_{ij0}^{\text{U.S.}}$ is the average export value of good j to the U.S. by firm i during 2013–2016 and j_i^e is the set of goods exported by firm i to the U.S. Following Benguria et al. (2022) and Chor and Li (2024), to avoid potential endogeneity issues stemming from reverse causality in firm's exports to the U.S. tariffs, we maintain export value weights $X_{ij0}^{\text{U.S.}}$ constant at the 2013–2016 average. Consequently, we have a fixed pre-trade-war weight alongside a time-varying, update-to-date tariff for each good from 2015 to 2022. We then calculate the weighted average for each good and aggregate these values to the firm level, which constitutes our measure of firm-level tariff exposure.

In a similar vein, on March 23, 2018, the Chinese government retaliated with a list of 128 products that would face 15%–25% tariffs

should the U.S.–China trade negotiations fail. We also construct firm i 's Chinese tariff exposure at time t : denoted as $\text{Tariff}_{it}^{\text{CHN}}$. We employ the natural logarithm of the tariff as the primary independent variable in our regression model. One concern with the current empirical setup is that focusing solely on the Sino-U.S. relationship may overlook other significant tariff relationships China has with regions such as the E.U. or Russia. To address this, we construct tariff exposure measures for these regions using the same methodology applied to U.S. tariff exposure for Chinese firms. These additional measures are incorporated into analysis.

3.2 | Firm's Integration of AI

We utilize the CSMAR database on firm digital transformation to construct an index measure for firm AI integration, extracting both quantitative indicators and textual information from annual reports. The index measures several key factors:

- Term frequency analysis of technical vocabulary (AI, machine learning, blockchain, cloud computing, big data) through structured textual examination.

- b. Capital allocation to AI-integrated projects (laboratories, platforms, facilities, equipment, infrastructure, information systems, smart interfaces, robotics).
- c. Executive leadership structure (inclusion of CIO, CTO, CDO positions).
- d. Strategic AI priorities in emerging technology, as reflected by the frequency of AI and digital transformation keywords in Management Discussion and Analysis (MD&A) sections.

Our methodology follows established textual analysis techniques for corporate disclosures (Ardito et al. 2024). The design mitigates endogeneity through the exogenous nature of U.S. tariff policy changes, which are external and unpredictable to firms. Tariff shocks precede firm responses such as AI integration, creating a clear temporal sequence. Year and industry fixed effects control for temporal trends, macroeconomic conditions, and sectoral heterogeneity, addressing reverse causality, simultaneity, and omitted variable concerns.

3.3 | Need for Primary Customer Engagement

We use three proxy variables to measure supplier firms' primary customer importance: overseas sales, asset specificity, and customer concentration.

- a. *Overseas sales*, measured as the percentage of international sales to total revenue, indicate the global reach of a supplier's customer base. Significant international sales reflect geographically diverse primary customers, increasing demand complexity due to varying market conditions (Johanson and Vahlne 2015).
- b. *Asset specificity*, measured as R&D expenses relative to annual sales, reflects investments dedicated to serving specific customers and indicates the firm's commitment to primary customer relationships (Hagedoorn and Duysters 2002; Rindfleisch and Heide 1997).
- c. *Customer concentration*, measured as the proportion of sales to major customers, indicates reliance on key customers. High concentration indicates that a substantial portion of revenue comes from a few primary customers, suggesting close and dependent relationships (Kale et al. 2002; Patatoukas 2012).

We use Principal Component Analysis (PCA) to combine these three proxies into a single composite measure.

3.4 | Supply Chain Performance

We use several variables, including the Bullwhip ratio, inventory levels, inventory turnover ratio, and inventory days, to assess supply chain performance. The Bullwhip ratio, defined as the volatility of production divided by the volatility of demand, is widely used to evaluate supply chain efficiency (Bayraktar et al. 2020). A ratio exceeding 1 indicates production volatility surpasses demand fluctuations, causing the bullwhip effect, where supply and demand mismatches escalate along the supply chain. Chinese exporters may experience heightened bullwhip

ratios due to unexpected increases in U.S. import tariffs, which disrupt demand and production alignment.

$$\text{Bullwhip Ratio} = \frac{V[\text{Production}]}{V[\text{Demand}]}$$

where the *production* is defined as $\text{production}_{it} = \text{demand}_{it} + \text{Inv}_{it} - \text{Inv}_{it-1}$, and demand_{it} is the sales revenue of firm i at year t , and Inv_{it} is the inventory of firm i at year t . We remove the time trend by applying logarithmic and first-difference transformations to both the production and demand series. Definitions of other, more straightforward supply chain performance measures are provided in Appendix 1.

3.5 | Baseline Regression Model

For hypothesis testing, we apply the following OLS regression model to examine the effect of firm tariff exposure on AI integration:

$$AI_{it} = \beta_0 + \beta_1 \text{Tariff}_{it}^{\text{U.S.}} + \beta_2 X_{it} + \text{year}_t + \text{ind}_j + \varepsilon_{it}$$

where i represents firms, and t represents years. $\text{Tariff}_{it}^{\text{U.S.}}$ is the firm i 's exposure to the U.S. tariff increase in year t . AI_{it} is the AI integration index of firm i in year t . Our firm-level measure of tariff exposure is time-varying and arguably exogenous. It assists us in identifying the causal impact of the sudden increase in the tariff on AI integration and the economic outcomes from both time series and cross-section.

X_{it} is a vector of control variables that may impact AI integration. Firm size (measured by log of total assets) captures the resource advantages larger organizations have for AI investments (Chen and Hambrick 1995). Financial position indicators include leverage (total debt to total assets), which may either constrain AI spending or motivate efficiency-seeking automation, and cash holdings (year-end cash to total assets), which enables firms to undertake substantial AI investments requiring significant upfront costs. For market-related factors, we include share turnover, as firms with higher liquidity often face investor pressure for innovation and can raise capital at lower costs to fund AI initiatives. We also consider structural characteristics through capital intensity (capital expenditure to total assets), recognizing that while capital-intensive firms may face integration challenges, they stand to benefit substantially from AI-driven automation. Ownership structure is captured through state ownership ratio (state-owned shares to total shares), as government involvement may slow AI adoption due to bureaucratic processes (Zhou et al. 2017). Finally, we include year (year_t) and industry (ind_j) fixed effects to account for temporal trends and sector-specific factors affecting both tariff changes and AI integration.

Figure 2 illustrates the evolution of our tariff exposure measure over time. The tariff exposure became noticeable only after 2018, aligning with the timeline of the trade war, which began in March 2018 when President Trump announced a 25% increase in steel and a 10% increase in aluminum import tariffs. In contrast, the tariff exposure to Russia and the E.U. remained essentially unchanged throughout the same period. Figure 3 shows how firms'

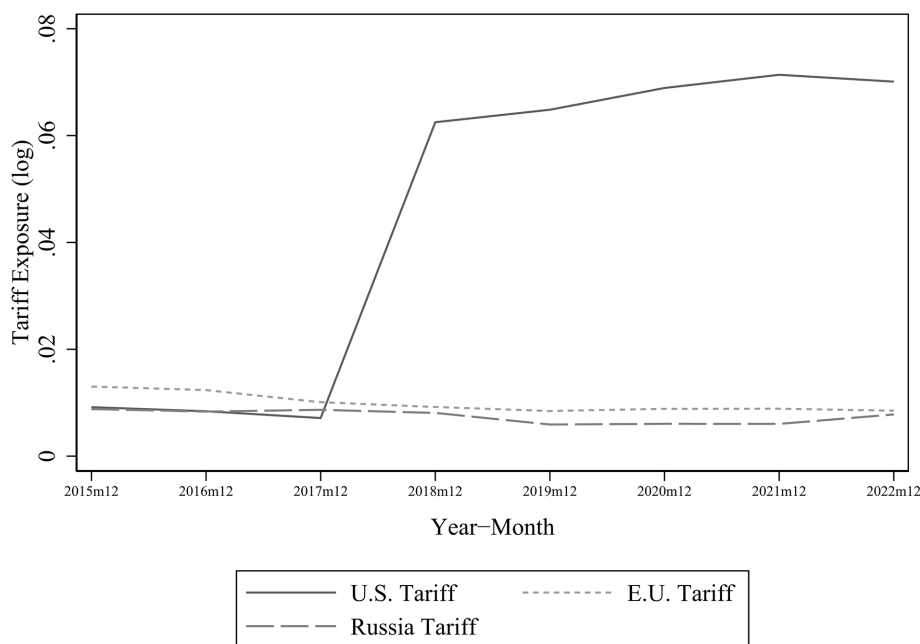


FIGURE 2 | Tariff exposure 2015–2022.

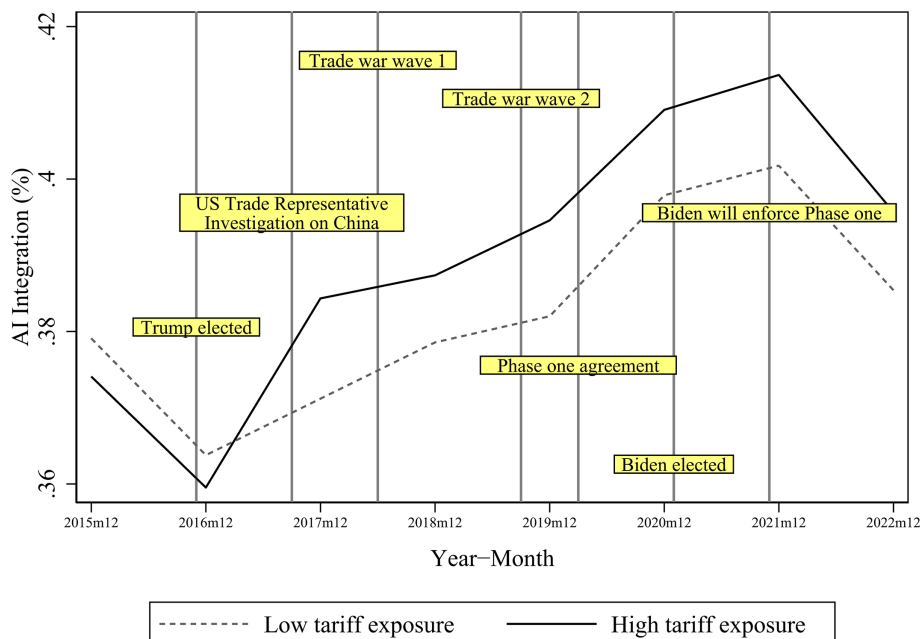


FIGURE 3 | AI integration and trade tension (2015–2022).

AI integration changes in response to the anticipation of tariff increases.

We divide our sample into high and low tariff exposure groups based on median exposure (0) and track annual AI integration. Before 2017, both groups showed similar AI integration patterns, with declines in 2015–2016. However, during Trump's presidency, following the August 2017 directive to investigate Chinese economic practices and the March 2018 tariff implementation, high-exposure firms significantly increased AI integration compared with low-exposure firms. This disparity continued after Biden's 2021 election, reflecting persistent trade policy uncertainty. Both groups showed declining AI integration

in 2022, potentially responding to reduced uncertainty under Biden's administration, including the late 2021 Phase One trade deal with China. Table 2 provides summary statistics, with detailed variable definitions in Appendix 1.

4 | Results

We present our results in three stages: first examining the direct effect of tariff exposure on AI integration, then investigating the moderating role of customer engagement need, and finally assessing the performance outcomes of AI integration. In Table 3, column 1 shows that a firm that has higher exposure

TABLE 2 | Summary statistics.

Variable name	Obs.	Mean	SD	Min	Median	Max
Tariff_US	17,904	0.058	0.123	0.000	0.000	0.983
Tariff_CN	17,904	0.050	0.101	0.000	0.000	0.700
Tariff_EU	17,904	0.011	0.022	0.000	0.000	0.182
Tariff_Rua	17,904	0.008	0.022	0.000	0.000	0.210
AI	17,640	0.390	0.110	0.214	0.374	0.800
Size	17,904	22.120	1.177	18.770	22.026	25.207
Turnover	17,624	4.313	2.953	0.236	3.541	13.426
Stateown	17,634	0.120	0.201	0.000	0.010	0.819
Leverage	17,904	0.411	0.250	0.008	0.395	11.386
Capex	17,904	0.054	0.062	−0.049	0.035	0.362
Cash	17,904	0.151	0.114	0.000	0.121	0.930
Customer_con	12,959	13.764	13.120	0.060	9.640	99.850
Asset_specificity	17,904	0.034	0.042	0.000	0.026	0.193
Sales_overseas	17,900	12.035	9.599	0.000	17.735	25.990
Sales	17,858	0.597	0.410	0.001	0.526	11.414
Tobin's Q	17,328	0.816	0.591	−0.833	0.730	6.891
Bullwhipratio	14,340	1.351	1.518	0.000	1.095	58.007
Inventory	17,665	−1.666	1.042	−14.698	−1.600	4.427
Inventory_turnover	17,454	6.180	12.173	0.322	3.388	96.072
Inventory_days	17,753	4.578	1.056	−9.283	4.629	10.354
Cluster	17,623	3.353	2.382	0.000	3.000	7.000
Railway_1933	17,623	11.134	6.983	0.000	11.000	21.000
Port_distance	16,420	0.051	0.0967	0.000	0.014	0.564

Note: See Appendix 1 for variable definitions.

to a U.S. tariff increase tends to integrate more AI technology as a response. The coefficient of the tariff exposure is positive and significant at a 1% level ($\beta = 0.042$; $p < 0.01$), suggesting that firms exposed more to the trade war are more involved in AI integration.

In column 2 of Table 3, we include firm exposure to Chinese import increases as a robustness check. Even after incorporating Chinese import tariffs and their interaction with U.S. tariffs, the coefficient for U.S. tariff exposure remains statistically positive and largely unchanged. This suggests that U.S. tariff exposure continues to significantly influence firms' AI integration, despite some moderation due to the effects of Chinese retaliatory tariffs. Additionally, the coefficients of control variables such as firm size and cash holdings show positive associations with AI integration, likely reflecting larger firms' greater capacity to invest in AI technologies. Conversely, state ownership and capital expenditure exhibit negative correlations with AI integration, possibly indicating that firms heavily invested in tangible assets may have fewer resources available for AI-related activities. Furthermore, coefficients for E.U. and Russia tariff exposure

show no statistical significance, consistent with findings illustrated in Figure 2, suggesting that AI integration primarily responds strategically to U.S. tariff changes rather than tariffs imposed by other countries.

Table 4 reports the results of the dynamics of how firms adopt AI technologies as a response to tariff change. We use year dummies to capture the effect of tariff exposure on AI integration. By setting the year 2018 as the benchmark, the coefficients of the other year dummies show relative magnitude with respect to that year. We note that the year dummies begin showing positive and significant effects only after 2018, thereby providing consistent evidence with our graphical analysis (Figure 3). Overall, the results in Tables 3 and 4 strongly support H1.

To investigate the moderating effect of the need for primary customer engagement on the relationship between tariff exposure and AI integration, we consider the interaction terms of the three proxy variables outlined in Section 3.3 with tariff exposure, respectively. According to column 1 in Table 5, H2 is supported: the interaction between overseas sales and tariff

TABLE 3 | The impact of tariff exposure on AI integration during US–China trade tensions.

	(1)	(2)	(3)
	AI	AI	AI
Tariff_US	0.042*** (3.91)	0.035** (2.41)	0.037** (2.34)
Tariff_CN		0.072*** (3.16)	0.073*** (3.10)
Tariff_US*Tariff_CN		−0.152 (−1.55)	−0.156 (−1.55)
Tariff_EU			−0.041 (−0.52)
Tariff_Rua			0.034 (0.48)
Stateown	−0.031*** (−4.49)	−0.030*** (−4.44)	−0.030*** (−4.43)
Leverage	0.000 (0.03)	0.000 (0.06)	0.000 (0.05)
Size	0.022*** (17.13)	0.021*** (16.60)	0.021*** (16.57)
Cash	0.019* (1.87)	0.020** (2.02)	0.020** (2.02)
Turnover	0.001*** (4.09)	0.001*** (3.87)	0.001*** (3.85)
Capex	−0.125*** (−8.09)	−0.124*** (−8.05)	−0.124*** (−8.04)
_cons	−0.169*** (−5.96)	−0.158*** (−5.55)	−0.158*** (−5.51)
ind	Yes	Yes	Yes
Year	Yes	Yes	Yes
N	17,623	17,623	17,623
R ²	0.506	0.507	0.507

Note: The dependent variable is AI integration. Tariff variables are natural logarithms of firm-level tariff exposure. Standard errors clustered by firm are in parentheses. *, **, *** indicate significance at 10%, 5%, and 1% levels. See Appendix 1 for variable definitions.

exposure is significantly positively related to the AI integration index ($\beta=0.002$; $p<0.05$). This suggests that especially for firms with higher uncertainty in their overseas demand, they would push harder to pursue AI integration when facing greater trade exposure. In column 2 of the same table, our finding also lends support to H2 as the interaction term between tariff exposure and asset specificity, as measured by the R&D expenses as a percentage of sales, shows up significantly positive ($\beta=1.059$; $p<0.01$). This indicates that the greater the asset specificity in terms of relationship-specific investment,

the more firms with higher trade exposure would be actively adopting an AI strategy.

In Table 5, Column 3, we find a significant positive interaction term between customer concentration, measured by the first major customer's sales revenue as a percentage of total sales, and tariff exposure ($\beta=0.004$; $p<0.01$). Again, our findings support H2; the positive effect of AI integration is reinforced in firms with higher customer concentration. The stronger the need for engagement with primary downstream customers, the more firms with a higher U.S. tariff exposure are involved in adopting AI technologies.

In addition, our findings indicate that the results remain consistent and robust when using the PCA-derived composite measure. Specifically, the interaction term between the first principal component and tariff exposure (CustEng_PC*Tariff_US), which captures the combined effect of the three dimensions, has a positively significant coefficient. This is consistent with the results of the separate customer engagement dimensions, suggesting that the overall need for a firm to enhance engagement with its primary customers significantly influences their integration of AI technologies, particularly in the face of intensified trade uncertainty.

Next, we investigate the impact of enhanced AI integration on firms' economic outcomes from a supply chain efficiency perspective. We introduce an interaction term between AI integration index and tariff exposure into an OLS model and employ a series of firm performance measures that capture supply chain efficiency, sales performance, and valuation as the dependent variable. Column 1 in Table 6 shows that more advanced AI integration enables firms that are more exposed to US tariffs to reduce the mismatch between production and demand, that is, a reduction in bullwhip ratio, with a statistically significant effect ($\beta=-1.774$; $p<0.1$). In Columns 2–4 of the same table, we further examine additional measures of supply chain performance, such as inventory level, inventory turnover ratio, and days of inventory turnover. Our findings indicate that a higher level of AI integration reduces inventory level ($\beta=-3.018$; $p<0.01$), accelerates inventory turnover ($\beta=28.123$; $p<0.05$) and shortens the average inventory turnover days ($\beta=-2.779$; $p<0.05$) for firms that are more exposed to U.S. tariff increases. Still in Table 6, Column 5 shows a higher level of AI integration mitigates the adverse impact of tariff exposure on firms' sales, that is, the interaction term coefficient is positively significant ($\beta=1.335$; $p<0.05$).

Moreover, Column 6 shows that for firms with higher tariff exposure, a higher AI integration level leads to a greater Tobin's Q ($\beta=0.977$; $p<0.05$), a measure of firm value and efficiency. The link between improved supply chain performance and a higher Tobin's Q is supported by several factors. Enhanced supply chain efficiency leads to cost savings, faster turnaround times, and improved inventory management, all of which contribute to better financial performance and higher market valuation. Superior supply chain performance also provides a competitive advantage, leading to increased investor confidence and higher market valuation. Additionally, effective supply chain management reduces operational risks and fosters stakeholder confidence, further enhancing Tobin's

TABLE 4 | Dynamics of tariff exposure and AI integration.

	(1)	(2)
	AI	AI
Year2015	0.000 (0.12)	0.001 (0.72)
Year2016	−0.013*** (−8.93)	−0.012*** (−7.72)
Year2017	0.001 (0.41)	0.002 (1.22)
Year2019	0.002 (1.62)	0.001 (0.46)
Year2020	0.009*** (5.94)	0.008*** (5.25)
Year2021	0.012*** (7.08)	0.011*** (6.44)
Year2022	−0.007*** (−4.35)	−0.008*** (−4.80)
Tariff_CN		0.062*** (3.95)
Tariff_EU		0.005 (0.06)
Tariff_Rua		0.044 (0.63)
Stateown	−0.032*** (−4.69)	−0.031*** (−4.48)
Leverage	−0.000 (−0.03)	0.000 (0.03)
Size	0.022*** (17.44)	0.022*** (16.62)
Cash	0.018* (1.77)	0.020* (1.96)
Turnover	0.001*** (4.21)	0.001*** (3.97)
Capex	−0.128*** (−8.28)	−0.125*** (−8.11)
_cons	−0.176*** (−6.22)	−0.161*** (−5.61)
ind	Yes	Yes
N	17,623	17,623
R ²	0.504	0.507

Note: The dependent variable is AI integration. Year dummies capture tariff exposure effects relative to benchmark year 2018. Standard errors clustered by firm in parentheses. *, **, *** indicate significance at 10%, 5%, and 1% levels. See Appendix 1 for variable definitions.

Q. The above tests provide strong and consistent support to H3. In all, AI integration during the trade war leads to an increase in supply chain performance, mitigating the adverse effects of the trade war.

To ensure the robustness and validity of our findings, we employ several approaches, focusing in particular on reverse causality and omitted variable bias. To address concerns that technologically oriented firms might be more likely to experience higher tariff exposure, which could create reverse causality, we first use lagged tariff exposure measures to establish temporal precedence (Table 7, Column 1). The results indicate that lagged U.S. tariff exposure has a positive and significant effect on AI integration ($\beta = 0.025$; $p < 0.05$), demonstrating that tariff exposure precedes firms' AI integration decisions.

Second, we conduct a subsample analysis comparing high-tech and non-tech firms (Table 7, Columns 2–3). The results reveal no significant difference in the tariff exposure effect between these two groups. A *t*-test comparing the coefficients of the two regressions shows no statistical difference (0.376), suggesting that our findings are not driven by differences in firms' technological orientation. These results collectively address concerns regarding reverse causality.

To account for potential omitted variables that may influence both tariff exposure and AI integration, we implement three distinct instrumental variable (IV) approaches.

First, we use the number of pre-WTO (before 2001) industrial clusters in each province as an instrument (Table 7, Columns 4–5). This instrument relies on the logic that historical industrial distribution impacts current tariff exposure but is unlikely to directly influence contemporary AI integration (Lu et al. 2019). The first-stage result indicates strong instrument relevance (*F*-statistic = 25.05), and the second-stage result confirms the positive and significant relationship between tariff exposure and AI integration. Second, we employ the density of railways in each province in 1933 as an instrumental variable (Table 7, Columns 6–7). This historical infrastructure measure predates current trade tensions, providing exogenous variation in modern trade patterns. The instrument demonstrates strong statistical power (*F*-statistic = 39.19), and the second-stage result aligns with our primary findings. Third, we use the distance between an exporting firm's city location and the nearest seaport as an instrument (Table 7, Columns 8–9). This approach leverages geographic location, which affects trade exposure but is plausibly exogenous to AI integration decisions. We calculate the inverse of the distance, where a larger value indicates closer proximity to the seaport. This instrument also proves strong (*F*-statistic = 25.59) and supports our main findings.

Across all three IV approaches, the second-stage results consistently show a positive and significant effect of tariff exposure on AI integration, further supporting the validity of our analysis and reinforcing our main findings. We conduct additional analyses to explore the quadratic relationship between tariff exposure and firm performance, moderated by AI integration. These analyses reveal important non-linear patterns that provide a more nuanced understanding of the relationship. The results, presented in Table 8, highlight how AI integration influences

TABLE 5 | Need for primary customer engagement, tariff exposure, and AI integration.

	(1)	(2)	(3)	(4)
	AI	AI	AI	AI
Sales_overseas*Tariff_US	0.002** (1.98)			
Sales_overseas	0.000 (0.94)			
Asset_specificity*Tariff_US		1.059*** (3.91)		
Asset_specificity		0.134*** (3.51)		
Customer_con*Tariff_US			0.004*** (5.60)	
Customer_con			−0.001*** (−6.98)	
CustEng_PC*Tariff_US				0.146*** (6.65)
CustEng_PC				−0.014*** (−4.77)
Tariff_US	−0.014 (−0.71)	−0.033* (−1.73)	−0.037** (−2.10)	0.004 (0.29)
Tariff_CN	0.050*** (2.95)	0.054*** (3.22)	0.065*** (3.39)	0.063*** (3.28)
Tariff_EU	−0.028 (−0.36)	0.019 (0.24)	0.029 (0.31)	0.076 (0.83)
Tariff_Rua	0.030 (0.43)	0.038 (0.55)	−0.056 (−0.70)	−0.035 (−0.45)
_cons	−0.146*** (−4.97)	−0.167*** (−5.83)	−0.105*** (−3.10)	−0.133*** (−3.92)
ind	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	17,619	17,623	12,717	12,714
R ²	0.507	0.510	0.515	0.513

Note: The dependent variable is AI integration. CustEng_PC is the first principal component of the three proxy variables (overseas sales, asset specificity, customer concentration). Standard errors clustered by firm are in parentheses. *, **, *** indicate significance at 10%, 5%, and 1% levels. See Appendix 1 for variable definitions.

the impact of tariff exposure on supply chain performance, sales performance, and firm value.

For supply chain performance, measured by the Bullwhip Effect, the linear interaction term between AI integration and tariff exposure is negative and significant ($\beta = -5.694$; $p < 0.05$). This indicates that AI integration helps mitigate supply chain inefficiencies caused by tariff exposure. However, the quadratic interaction term

is positive but insignificant, suggesting that while AI integration initially reduces supply chain inefficiencies, the benefit may diminish at higher levels of tariff exposure.

For sales performance, the results show a positive and significant linear interaction effect ($\beta = 2.583$; $p < 0.10$), indicating that AI integration supports firms in maintaining sales performance under tariff pressure. However, the quadratic interaction term

TABLE 6 | Tariff exposure, AI integration, and firm performance.

	(1)	(2)	(3)	(4)	(5)	(6)
	Bullwhip	Inventory	Inv. T.O.	Inv. days	Sales	Tobin's Q
AI*Tariff_US	−1.744*	−3.018***	28.123**	−2.779**	1.335**	0.977**
	(−1.72)	(−2.97)	(2.32)	(−2.33)	(2.52)	(2.02)
Tariff_US	0.422	1.325***	−11.928***	1.115**	−0.542**	−0.411*
	(0.89)	(3.16)	(−2.62)	(2.37)	(−2.37)	(−1.94)
AI	0.262	0.482**	−7.588***	0.723***	−0.146**	0.021
	(1.43)	(2.45)	(−3.55)	(3.55)	(−2.16)	(0.25)
Stateown	0.174*	−0.031	2.771**	−0.211**	0.113***	−0.165***
	(1.96)	(−0.35)	(2.40)	(−2.28)	(2.78)	(−4.27)
Leverage	0.094	0.106*	1.011	−0.078	0.136***	−0.078
	(1.55)	(1.71)	(1.50)	(−0.89)	(2.78)	(−1.62)
Size	−0.026	−0.058***	0.228	−0.079***	0.018**	−0.229***
	(−1.56)	(−3.74)	(1.27)	(−4.58)	(2.30)	(−26.93)
Cash	−0.073	−1.215***	5.707***	−0.762***	0.201***	0.891***
	(−0.51)	(−8.10)	(2.72)	(−4.61)	(2.78)	(12.49)
Turnover	0.012*	0.006	0.020	−0.003	−0.000	−0.004**
	(1.70)	(1.62)	(0.43)	(−0.73)	(−0.26)	(−2.17)
Capex	−0.313	−1.275***	6.548**	−1.585***	−0.084	1.213***
	(−1.21)	(−6.09)	(2.29)	(−7.68)	(−1.06)	(13.29)
Tariff_CN	0.329	0.221	−2.880**	0.106	0.061	−0.134
	(1.30)	(1.53)	(−2.55)	(0.68)	(0.76)	(−1.53)
Tariff_EU	−0.016	−1.005	1.768	−1.825**	0.892	0.258
	(−0.02)	(−1.24)	(0.25)	(−2.08)	(1.25)	(0.65)
Tariff_Rua	0.327	0.174	−3.823	0.501	−0.525	0.028
	(0.42)	(0.28)	(−0.79)	(0.76)	(−1.14)	(0.08)
_cons	1.850***	−0.076	−0.260	6.750***	0.276*	6.337***
	(5.16)	(−0.23)	(−0.07)	(18.10)	(1.70)	(35.68)
ind	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
N	14,324	17,427	17,437	17,488	17,612	17,322
R ²	0.008	0.167	0.136	0.165	0.075	0.457

Note: The dependent variable is performance measures: bullwhip ratio (Column 1), inventory (Column 2), inventory turnover ratio (Column 3), inventory days (Column 4), sales (Column 5), and Tobin's Q (Column 6). Standard errors clustered by firm in parentheses. *, **, *** indicate significance at 10%, 5%, and 1% levels. See Appendix 1 for variable definitions.

is negative but insignificant, suggesting that the positive impact of AI integration on sales performance may plateau at extreme levels of tariff exposure.

The results are most pronounced for firm value, measured by Tobin's Q. A strongly significant positive linear interaction term is observed ($\beta = 4.170$; $p < 0.01$), indicating that AI integration enhances firm value with increasing tariff exposure. However, the

quadratic interaction term is negative and significant ($\beta = -9.862$; $p < 0.01$), revealing an inverted U-shaped relationship. This indicates that while AI integration initially helps firms create value in response to tariff exposure, its benefits diminish and eventually decline at extreme levels of exposure. This finding suggests the existence of an optimal level of AI investment relative to tariff exposure, beyond which additional AI integration cannot fully offset the adverse effects of extreme tariff exposure.

TABLE 7 | Robustness check.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	AI	AI	AI	Tariff_US	AI	Tariff_US	AI	Tariff_US	AI
L. Tariff_US	0.025** (2.08)								
Tariff_US		0.043*** (2.95)	0.034* (1.80)		0.854*** (3.47)		0.389** (2.50)		1.986*** (7.09)
Cluster				0.003*** (5.21)					
Railway_1933						0.001*** (5.74)			
Port_distance								0.041** (2.36)	
_cons	−0.182*** (−6.14)	−0.247*** (−4.87)	−0.122*** (−3.14)	−0.172*** (−4.83)		−0.105*** (−3.01)		−0.174*** (−4.55)	0.116** (2.06)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ind	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>t</i> -test		0.376							
Cragg–Donald Wald <i>F</i> statistic				25.05		39.19		25.59	
<i>N</i>	14,837	9433	8190	17,623	17,623	17,623	17,623	16,420	16,420
<i>R</i> ²	0.511	0.588	0.226	0.133	−0.655	0.133	−0.052	0.135	0.567

Note: Column 1 uses lagged tariff exposure. Columns 2–3 compare high-tech vs. non-tech firms. Columns 4–9 use instrumental variables: industrial clusters pre-2001 (Columns 4–5), railway density in 1933 (Columns 6–7), and distance to nearest seaport (Columns 8–9). Standard errors clustered by firm in parentheses. *, **, *** indicate significance at 10%, 5%, and 1% levels.

5 | Discussion and Conclusions

This study examines how firms integrate AI in response to trade tariff disruptions and how customer engagement needs influence this response. We extend induced innovation theory to develop adversity-induced innovation theory, demonstrating that geopolitical disruptions drive technological change beyond traditional factor price mechanisms. Our findings reveal three key insights: tariff exposure induces AI integration as a compensatory innovation strategy, the need for primary customer engagement moderates this relationship, and AI integration improves firm performance up to an optimal threshold beyond which extreme tariff exposure overwhelms technological adaptation. These results advance the understanding of innovation under adversity and establish boundary conditions for when and how firms leverage technology to address external shocks.

5.1 | Theoretical Contributions

This study advances innovation theory by reconceptualizing induced innovation as a response to adversity beyond traditional

factor price changes. Three primary theoretical contributions follow from our empirical findings. First, we extend induced innovation theory from its traditional focus on factor price signals to encompass geopolitical disruptions as innovation drivers. Traditional induced innovation theory explains how rising factor costs (labor, energy) direct technological change toward economizing on expensive inputs (Acemoglu 2010; Hicks 1932; Popp 2002; Fleisher et al. 2021). Our findings demonstrate that this logic extends to geopolitical adversities: tariff exposure functions as an economic shock that induces compensatory innovation through AI integration. This represents a theoretical advance because it identifies adversity as a distinct category of innovation inducement, separate from but related to price-based mechanisms. The theory now explains innovation responses to sudden external shocks (tariffs, trade disruptions) rather than only gradual market-driven price adjustments. We term this extension adversity-induced innovation theory, broadening the scope of conditions under which firms pursue technological solutions.

Second, our findings identify customer engagement as a critical boundary condition that determines when adversity induces innovation. The need for primary customer engagement

TABLE 8 | Quadratic relationship of tariff exposure and firm performance.

	(1) Bullwhip	(2) Sales	(3) Tobin's Q
AI*Tariff_US	−5.694** (−2.04)	2.583* (1.89)	4.170*** (3.34)
AI*Tariff_US_SQT	12.843 (1.60)	−4.185 (−1.08)	−9.862*** (−3.04)
AI	0.300 (1.62)	−0.149** (−2.16)	−0.011 (−0.13)
Tariff_US	2.374* (1.77)	−1.199* (−1.78)	−1.728*** (−3.00)
Tariff_US_SQT	−6.060 (−1.50)	2.248 (1.16)	4.012** (2.53)
_cons	1.782*** (4.85)	0.263 (1.64)	6.373*** (36.06)
Controls	Yes	Yes	Yes
ind	Yes	Yes	Yes
Year	Yes	Yes	Yes
N	14,324	17,612	17,322
R ²	0.008	0.075	0.457

Note: Quadratic relationship between tariff exposure and firm performance. Dependent variables: bullwhip ratio (Column 1), sales (Column 2), and Tobin's Q (Column 3). Main variables are AI integration interacted with tariff exposure (linear and quadratic terms). Standard errors clustered by firm in parentheses. *, **, *** indicate significance at 10%, 5%, and 1% levels.

acts as a moderating mechanism that strengthens the relationship between tariff exposure and AI integration. Firms with higher customer engagement needs (measured through overseas sales, asset specificity, and customer concentration) respond more strongly to tariff exposure through AI integration. This finding advances stakeholder engagement theory (Hollebeek et al. 2022) by demonstrating that stakeholder relationships do not merely benefit from technological innovation but actively shape the innovation response itself. Customer engagement creates both motivation (understanding customer needs under tariff pressure) and capability (collaborative problem-solving) that strengthen the adversity–innovation link. This mechanism was absent from traditional induced innovation frameworks, which treated firms as uniform responders to price signals.

Third, we provide empirical evidence of the performance consequences of adversity-induced AI integration. Our results demonstrate that tariff-exposed firms improve supply chain efficiency, sales performance, and market value through AI integration. The inverted U-shaped relationship between tariff exposure and firm value reveals theoretical boundaries: AI integration effectively mitigates moderate tariff exposure but cannot fully offset extreme disruptions. This finding contributes to AI adoption literature (Ardito et al. 2024; Rammer et al. 2022) by establishing

the conditions under which AI delivers strategic value, moving beyond adoption studies to performance outcomes. The non-linear relationship suggests that adversity-induced innovation has optimal ranges, beyond which the compensatory benefits diminish.

5.2 | Managerial Implications

This research provides actionable insights for businesses facing external shocks, particularly trade tariffs. First, managers should consider AI as a central tool for strategic realignment and supplier-customer engagement. Our findings demonstrate that AI integration improves supply chain efficiency, optimizes sales performance, and strengthens stakeholder relationships even amid geopolitical challenges. AI facilitates value co-creation, allowing suppliers and customers to collaborate on innovation and increase the overall value derived from their interactions. These results suggest that companies can use AI to manage tariff-induced uncertainty rather than relying solely on traditional reactive measures.

Second, managers should recognize adversity as an opportunity for compensatory innovation. In the face of challenges such as trade wars, companies can identify opportunities to advance technologies, refine resource allocation, and adjust production techniques. Implementing AI in operations enables these compensatory innovations, allowing firms not only to survive but to strengthen their competitive position during external shocks.

Third, companies should prioritize AI-driven initiatives that address stakeholder needs, particularly customers affected by external shocks. AI-enabled customer insights can support the development of personalized products and services, fostering deeper supplier–customer engagement. Rather than treating AI as a temporary solution, firms should integrate AI into core business processes to achieve sustainable benefits, emphasizing stakeholder-oriented value co-creation. The study highlights that AI is a strategic resource capable of driving innovation, engagement, and resilience, and organizations should actively embed AI in their strategies to maximize these benefits.

5.3 | Limitations and Future Research

Our research has several limitations that suggest directions for future studies. First, our focus on customer engagement excludes other stakeholder groups. Future research could examine how AI strategies affect government bodies, employees, and other stakeholders to enhance strategic decision-making. Second, this study centers on US–China trade tensions. Further research should explore how firms employ AI during health crises, economic downturns, or policy changes to test the extension of adversity-induced innovation theory in different contexts. Third, our measurement approach has constraints. Although we quantify AI integration through investments and projects, case studies or interviews could provide richer insights into how AI influences innovation, stakeholder engagement, and firm stability. Finally, our sample is limited to publicly listed Chinese firms, which may restrict generalizability. Future studies should investigate firms in diverse markets and ownership structures

to assess how advanced technologies are leveraged under trade tensions across varied contexts.

Ethics Statement

The authors have read and agreed to the Committee on Publication Ethics (COPE) international standards for authors.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data are not publicly available due to privacy or ethical restrictions. The data that support the findings of this study are available on reasonable request from the corresponding author.

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Appendix 1

Variable Definitions

Variables	Symbols	Definitions
Dependent variables	AI	The AI Integration Index is a multi-dimensional measure of AI innovation in Chinese-listed firms, derived from annual reports. It includes digital technology terms (e.g., AI, machine learning, blockchain, and cloud computing), AI-related investments, key managerial roles (CIO, CTO, CDO), strategic initiatives, patents, product launches, process improvements, and external factors like industry R&D and regional infrastructure.
	Sales	Sales revenue as a percentage of total assets.
	Tobin's Q	The firm's market value divided by the book value of total assets.
	Bullwhipratio	The bullwhip ratio.
	Inventory	The firm's inventory divided by the sales in year t (log).
	Inventory_turnover	The operating costs are divided by the inventory at the end of year t .
	Inventory_days	The natural logarithm of 365 divided by Inventory_turnover.
Explanatory variables	Tariff_US	Natural log of the firm-level exposure to the U.S. tariff increase.
	Tariff_CN	Natural logarithm of firm-level exposure to Chinese tariff increase.
	Tariff_EU	Natural log of the firm-level exposure to the E.U. tariff increase.
	Tariff_Rua	Natural log of the firm-level exposure to the Russia tariff increase.
	Size	Natural logarithm of the firm's total assets.
	Turnover	The total number of shares traded divided by the total number of shares outstanding in year t .
	Stateown	The number of state-owned shares divided by the number of total shares.
	Leverage	The total liabilities divided by total assets in year t .
	Capex	The firm's capital expenditure in year t .
	Cash	The ratio of cash holdings (year-end) to total assets in year t .

Variables	Symbols	Definitions
Moderation variables	Sales_overseas	The natural logarithm of overseas sales.
	Asset_specificity	The R&D expenses as a percentage of sales revenue.
	Customer_con	The first major customer's sales revenue as a percentage of total sales revenue.
	CustEng_PC	The overall customer engagement moderator is computed as the 1st principal component of the three individual proxy variables.
Instrumental variables	Cluster	The number of industrial clusters in each province before 2001.
	Railway_1933	The number of cities which have railways in the province in the year 1933.
	Port_distance	The inverse of the distance (in km) between an exporting firm's city location and the nearest seaport.

Biographies

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